

BRIEF 01

Heart and Arteries

Two human studies, two distinct questions

VERIFIED PRIMARY STUDIES LINKED

Human studies have detected plastic-derived material in diseased artery plaque and coronary blood. These findings are important associations; they do not prove that plastic directly causes heart attacks or strokes.

WHAT THE CURRENT MATERIAL SUPPORTS

- A 2024 human study detected microplastics and nanoplastics in more than half of surgically removed carotid plaques.
- Plastic-containing plaques were associated with a higher combined rate of heart attack, stroke, or death during follow-up.
- A 2026 coronary-blood study reported detection in 84% of acute-heart-attack patients, 40% with chronic coronary disease, and 32% with normal coronary arteries.
- Higher particle measurements were reported alongside higher inflammatory markers in that study.

WHAT IT DOES NOT ESTABLISH

- The studies do not establish that particles caused the cardiovascular events.
- They do not support routine clinical microplastic testing or a new treatment pathway.
- Established risk factors - blood pressure, cholesterol, diabetes, smoking, obesity, and inactivity - remain primary prevention priorities.

INTERVIEW-SAFE LANGUAGE	Say: 'The studies found plastic-derived material and a concerning association that deserves further investigation.' Avoid: 'Microplastics have been proven to cause heart attacks.'
SOURCE STATUS	Primary sources linked on the site: New England Journal of Medicine (2024) and European Heart Journal (2026). Source draft: The heart.pdf.

FULL TOPIC AND SOURCE RECORD

saynotoplastic.com/science/body/cardiovascular-system

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